

Where data lives

Operational against analytical systems, and your first BigQuery query.

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RETRIEV
AL

Yesterday's question, finished

Where is the best falafel in Amman? We stopped at the blueprint.

The winner depends on what you count

Most reviews

The busiest place in the city. Thousands of people have been, and said something afterwards.

This measures how many people went. That is closer to popularity, or to being on a main road, than it is to quality.

Highest rating

A small place with forty reviews and an average of 4.9.

This measures how much the people who went liked it. Forty opinions is a thin basis for a claim about a city.

What this answer cannot carry

01 **Only people who wrote a review are counted**

Everyone who ate and said nothing is missing, and they are not a random group.

02 **One row is a place, not a person**

Counting rows counts shops. It does not count how many people prefer each one.

03 **A 4.9 from forty and a 4.1 from five thousand**

These are not the same claim, and averaging them together hides the difference.

The careful answer, and the honest one

The careful answer

There is no single best. There is a best on rating, a best on review count, and they are usually different places.

Report rating and review count separately, name the district, and say how many reviews sit behind each number.

The honest answer

Name one place. Say which measure it wins on, and say where it would lose.

You only get the second answer once you can show the first.

Every problem on that list was a question about what one row means. Today we look at where those rows come from.

What we cover today

- 01 **Where data lives**
- 02 Getting it out, and getting it ready
- 03 Why the cloud warehouse exists
- 04 The pipeline, and where SQL sits on it

PART 01

Where data lives

Files, systems, and the difference between running a business and asking questions about it.

START HERE

You have already worked with data

A spreadsheet on your laptop. A CSV somebody emailed you. A PDF bank statement. The photographs on your phone. Every one of those is data, and most of you have opened all four this month.

The job is not learning what data is. It is learning where a business keeps it, which is almost never a file on somebody's laptop.

	A	B	C	D	E	F
1	claim_id	state	claim_type	reported_date	amount_usd	status
2	CLM-1001	CT	auto	1/4/26	2450	closed
3	CLM-1002	MA	property	1/7/26	18200.5	open
4	CLM-1003	CT	auto	1/9/26	975.25	closed
5	CLM-1004	NY	auto	1/11/26	5600	open
6	CLM-1005	CT	property	1/15/26	42100	open
7	CLM-1006	RI	auto	1/18/26	1320.75	closed
8	CLM-1007	MA	property	1/22/26	8750	closed
9	CLM-1008	CT	auto	1/25/26	3010.4	open
10	CLM-1009	NY	property	1/29/26	15400	open
11	CLM-1010	CT	auto	2/2/26	640	closed
12	CLM-1011	VT	auto	2/5/26	2890.1	open
13	CLM-1012	MA	auto	2/9/26	4125	closed

FILES

The formats you will be handed

Format	What it holds	Where you meet it
CSV	Rows and columns, plain text, no types	Exports from almost any system
Excel	Rows, columns, formulas, several sheets	Finance teams, and everyone else
JSON	Nested records that carry their own labels	Anything that came from an API
Parquet	Columns stored separately, compressed	Data lakes, and why they are fast
PDF	A picture of a table, not a table	Price lists, reports, invoices

XML behaves like JSON for our purposes. Parquet comes back after the break, when we get to storage.



Volume

How much there is,
and whether one
machine can hold it



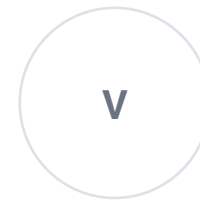
Velocity

How fast it arrives,
and whether you
can keep up



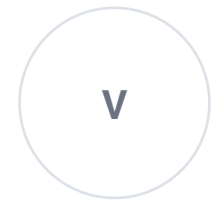
Variety

How many shapes,
and whether they
fit one table



Veracity

Whether you
can trust it



Value

Whether it is worth
keeping at all

Laney, META Group, 6 February 2001

added later, by other people

Three of these were written before anyone said the words “big data”.

— Volume, velocity and variety. Veracity and value were added later, by other people.

Where each of those lands today

Variety: Data comes in different formats

Tables, text, images, video. Only tables fit neatly into SQL

Velocity: Data arrives at different speed

Some systems send updates daily, others every second. All of it has to land somewhere first.

Veracity: Data is often incomplete or wrong

Duplicate customers, blank fields, typos. The same data quality issues we've always had.

Volume: Data outgrew a single machine

Once data got too big for one server, storage had to move to distributed systems and the cloud.

Nakheel Retail

8 shops
1 website
about 400 orders a day

a worked example, used all afternoon

01 **POS, point of sale**
one till per shop, on-premise

02 **Website order database**
cloud-hosted

03 **Finance spreadsheets**
monthly, on a shared drive

04 **Delivery partner CSV**
emailed every week

05 **Customer service email**
free text, never stops

06 **Website clickstream**
JSON, from an analytics tag

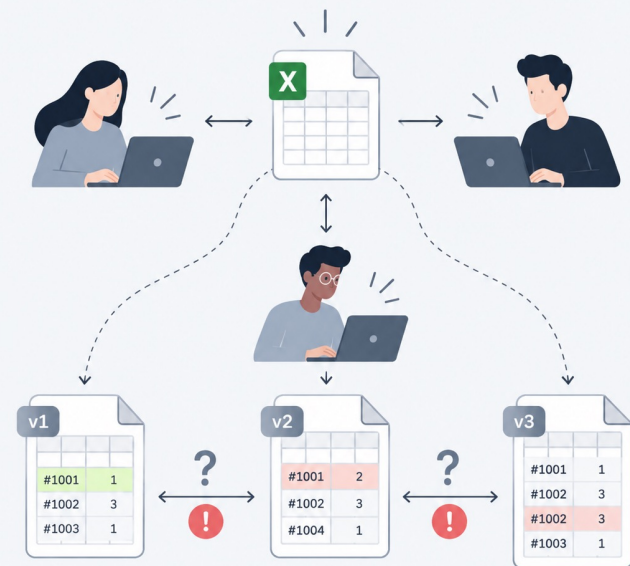
Six sources. Six owners. Six different arrival times. None of them agree with each other yet.

FILES

Why files stop working

Two people open the same spreadsheet and both change it. Which copy is right? Now do that across eight shops, four hundred orders a day, and a website that never closes.

A file has no answer to that question. A database does, and settling it is the whole reason databases were invented.

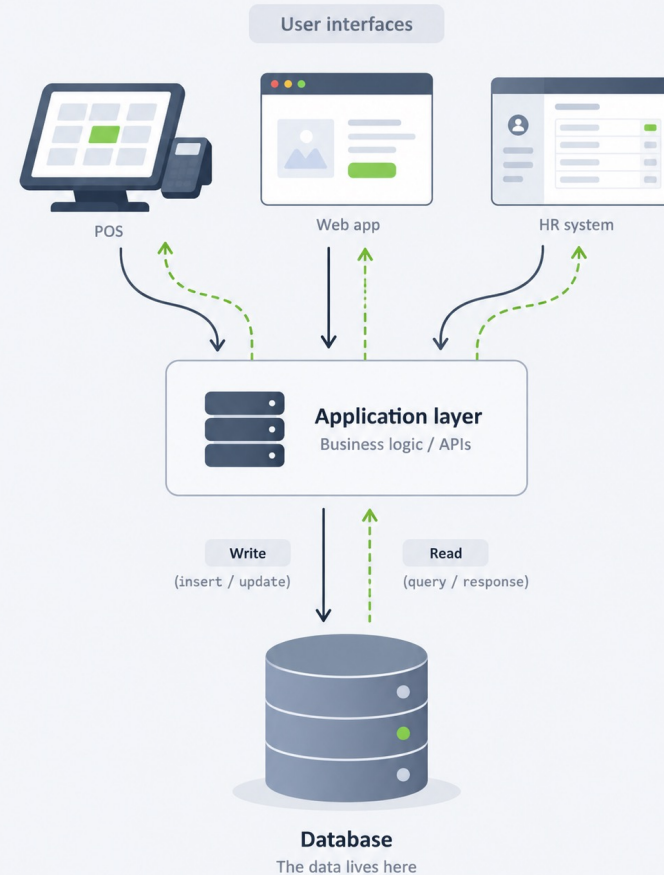


SYSTEMS

So the data lives in a system instead

The POS in each shop. The website that takes orders. The HR system that holds your contract. Each one is software sitting on top of a database, writing rows for people all day.

Nobody types into these databases directly. They are written to by an application, one small change at a time, thousands of times a day.



A machine you own, or one you rent

On-premise

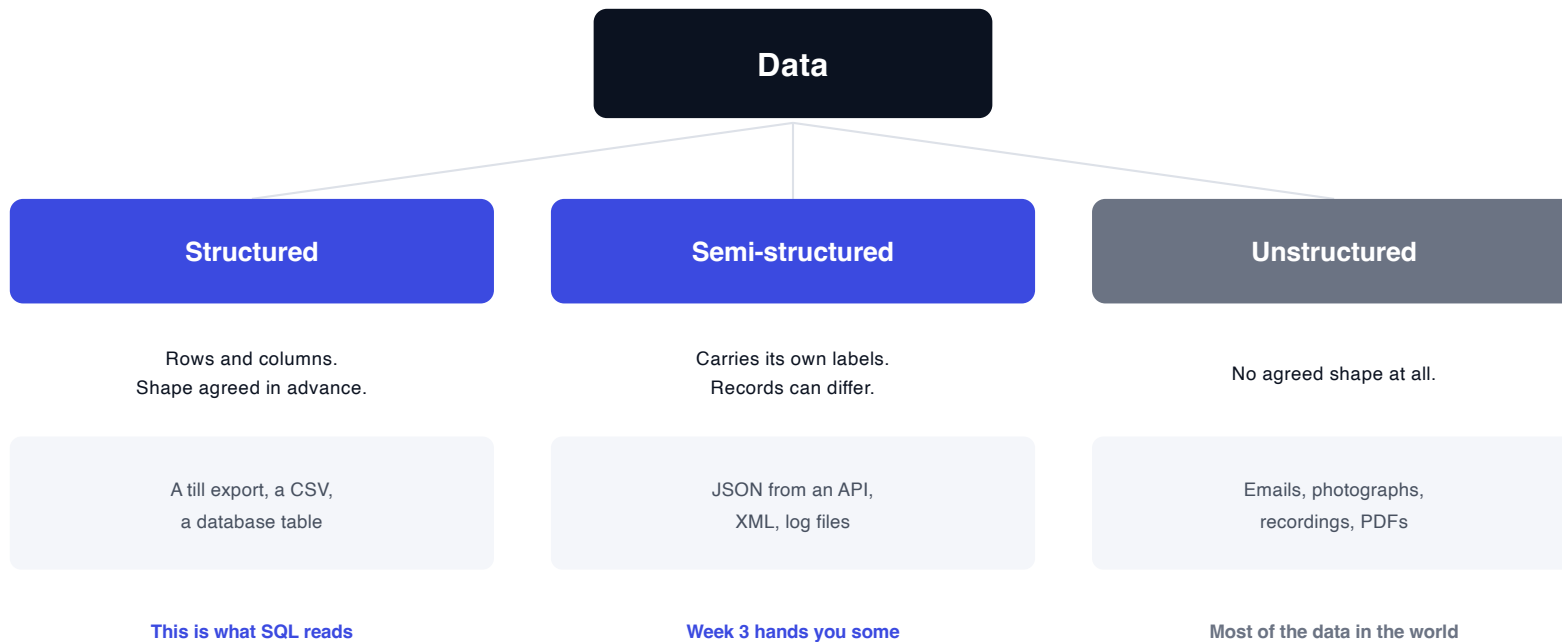
A server in your building. You bought it, you house it, you cool it, you patch it, and you chose its size three years ago based on a guess.

Still normal in Jordanian banking, government and healthcare, for reasons we come to after the break.

Cloud

A server in somebody else's building. You rent it, they patch it, and you change how much you are using on a Tuesday afternoon.

The same database software runs in either place. Where it sits is a separate decision from what it is.



Almost everything a BI analyst is asked for comes out of the first box.

— Structured, semi-structured, unstructured. The first one is what SQL reads.

OLTP

The system that runs the business

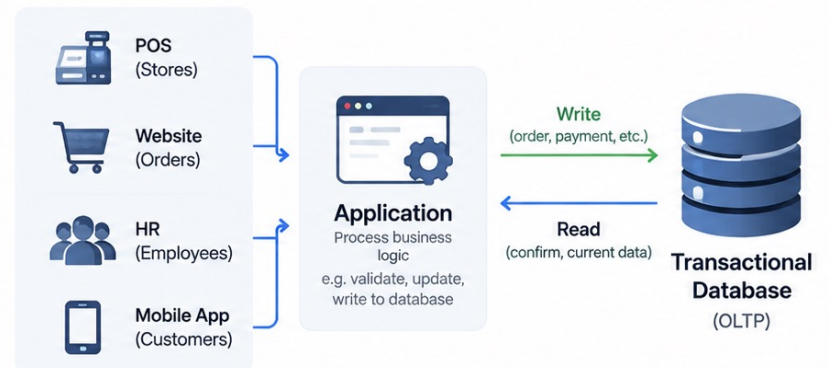
A customer pays at 14:32. Four rows get written: one order and three items. It takes under a second, it happens for one customer, and nothing else in the shop is touched.

Thousands of tiny writes a day, each one correct, each one fast, and nobody ever asks it what happened last month. This is what people mean by OLTP.

OLTP

Record what is happening now

Many users. Many small transactions. Fast. Accurate.



Example: A customer pays at 14:32. One order and three items are written to the database in under a second.

- Millions of small writes
- Real-time updates
- Keeps the business running

OLAP

The system that answers questions

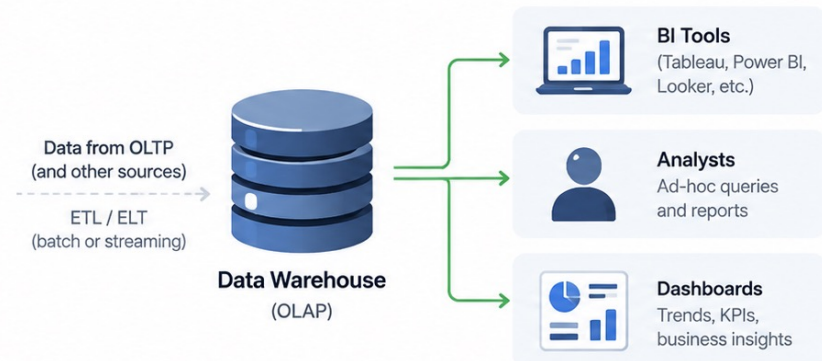
Which category made the most revenue last month? That one question reads every order in the period, adds them up by category, and sorts the result.

One user, one answer, a very large read, and nobody is standing at a till waiting for it to finish. A completely different job from the one the till is doing.

OLAP

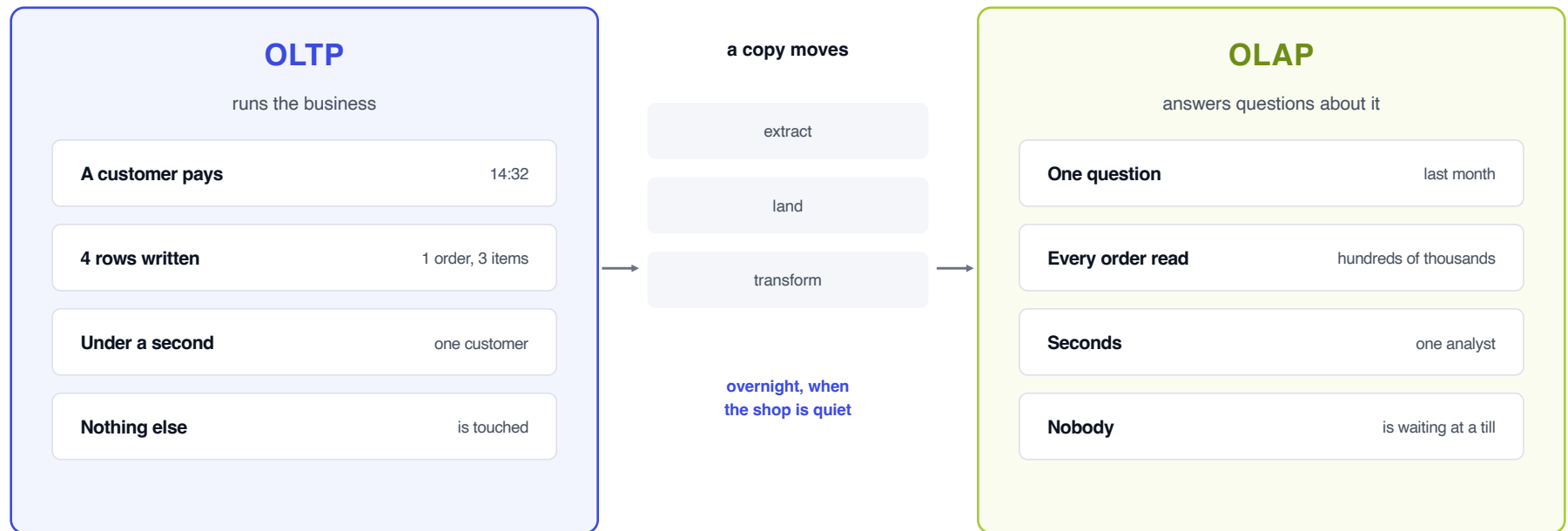
Understand what happened

Fewer users. Large reads. Complex queries. For analysis and insights.



Example: Which category made the most revenue last month? The query reads millions of rows, aggregates the data, and returns the result.

- Large, complex reads
- Historical data and aggregations
- Supports analysis and decision making



Same company, same data. One takes the order, the other explains what they add up to.

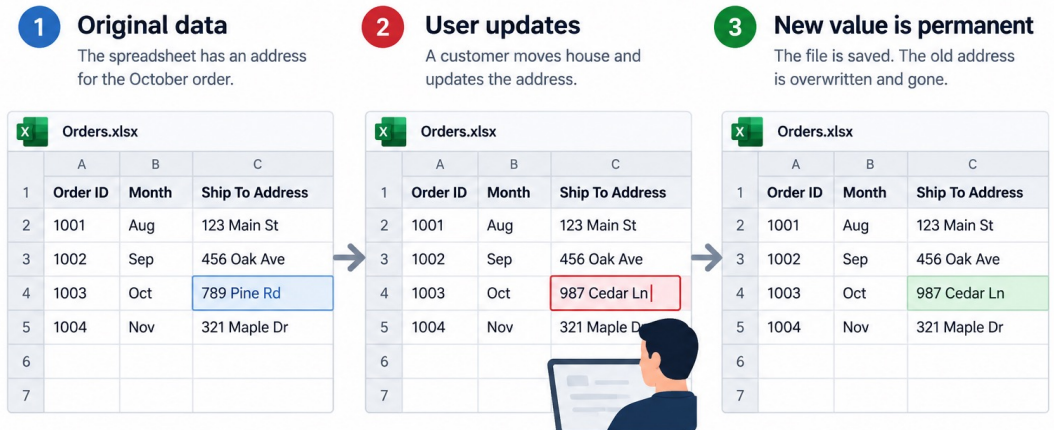
— Same company, same data. One takes the order, the other explains what they add up to.

HISTORY

What is true now (As-Is vs As-Was)

A customer moves house and updates their address on the website. The application overwrites the old address, which is exactly the right thing for it to do.

Now answer this: which address did the October order ship to? It is gone, and nothing went wrong. The operational system was never asked to remember.



Where the data ends up sitting

Database (think OLTP)

Rows, written by an application, one at a time.
Shaped for running the business. This is the till.

Data warehouse

Cleaned, organised, query-ready tables. Shaped for answering questions. This is where you will work.

Data lake

Cheap storage for files of any shape, kept as they arrived, including things nobody has opened in years.

Lakehouse

A lake you can query like a warehouse. Recognise the word; you are not being tested on it.

STORAGE

The names you will see on job adverts

Kind	Products you will see named
Operational database	PostgreSQL, MySQL, SQL Server, Oracle
Data warehouse	BigQuery, Snowflake, Redshift, Synapse
Data lake	Google Cloud Storage, Amazon S3, Azure Data Lake Storage
Lakehouse	Databricks, Microsoft Fabric

You are not tested on this list, and neither is PL-300. It is here so a job advert stops looking like a foreign language.

PART 02

Getting it out, and getting it ready

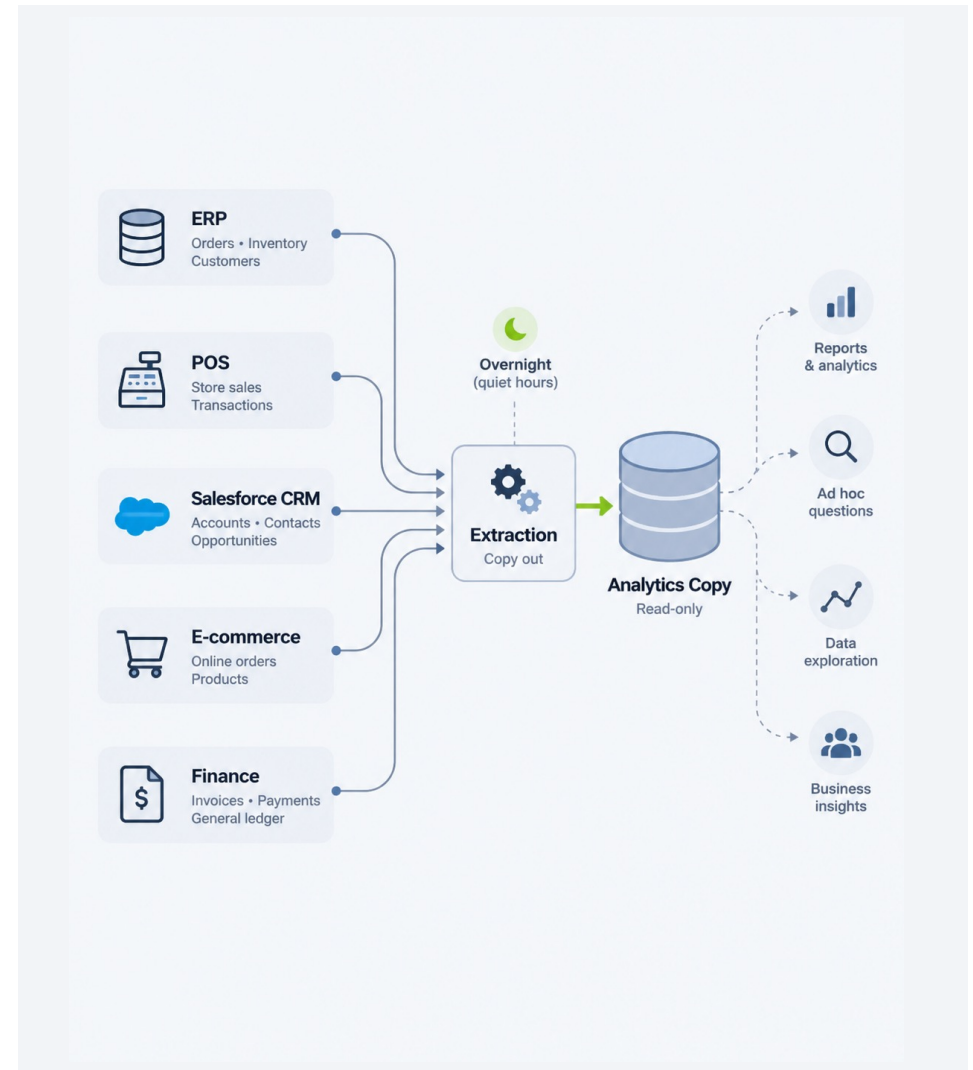
Why we take a copy, where we put it, and what happens before anyone can trust a number.

EXTRACTION

Why we take a copy out

Their system runs the business. Your question reads every order for a month. Run that against the till database at six on a Thursday and the shop slows down while you wait.

So we take one copy out, overnight while the shop is quiet, and ask every question of the copy. They feel the copy being taken once. They never feel the questions at all.



How it gets from them to you

Somebody sends you a file

The delivery partner emails a CSV every Sunday.
Simple, and completely outside your control.

You download an export

You log in and click export. Fine once, painful every Monday morning for a year.

You connect directly

A live connection to the database or the warehouse. This is what Power BI does in Module 3.

You call an API

You ask a service for data and it answers, usually in JSON. This is the semi-structured box.

How fresh can you honestly promise

Push or pull

Does the data arrive on its own, or do you go and fetch it? If somebody else pushes it to you, their schedule is now your schedule.

The Sunday CSV means nothing downstream of it can ever be fresher than a week old.

Batch or streaming

Does it come once a night in one lump, or continuously as things happen? Batch is far more common and far easier to reason about.

Almost everything in this course is batch. Streaming is a real thing and a different job.



one
place

You cannot clean and join six things that arrive like this. Catch them first, exactly as they come.

— Six systems, six arrival patterns, one day. This is why a landing zone exists.

ONE PLACE, FOUR NAMES

Landing, staging, raw, bronze

Four words for the same thing. Older systems said staging or landing. People working with lakes and lakehouses say raw, or bronze. You will hear all four, sometimes in the same meeting.

The rule underneath every one of those names is identical: store it exactly as it arrived and change nothing, so you can always go back to what the source actually sent.



PAUSE

Is raw data **useful**?

Take two or three answers before you advance. There is no trick in the question.

RAW

This is what raw looks like

order_id	status	ship_partner	city
1001	completed	aramex	Amman
1002	Completed	Aramex	amman
1003	COMPLETE	ARAMEX	
1004	cancelled	Aramex	Amman
1004	cancelled	Aramex	Amman

Four spellings of one company. Three of one status. A blank city, a trailing space, and order 1004 twice.

What a transformation is

01 **Rename.** Eight tills call the same field eight things. Pick one name.

02 **Re-type.** A date stored as text cannot be sorted or filtered as a date.

03 **Filter.** Drop the rows this question has no business counting.

04 **Derive.** Build a new column out of ones you already have.

05 **Join.** Put two tables side by side on a shared key.

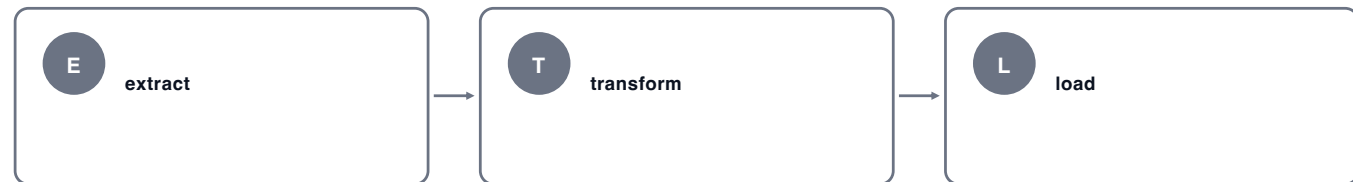
06 **Aggregate.** Fold many rows into one, with a total or a count.

07 **Fill or Drop.** A blank field either gets a default value or the row gets removed.

08 **Dedupe.** Two rows describing the same thing collapse into one.

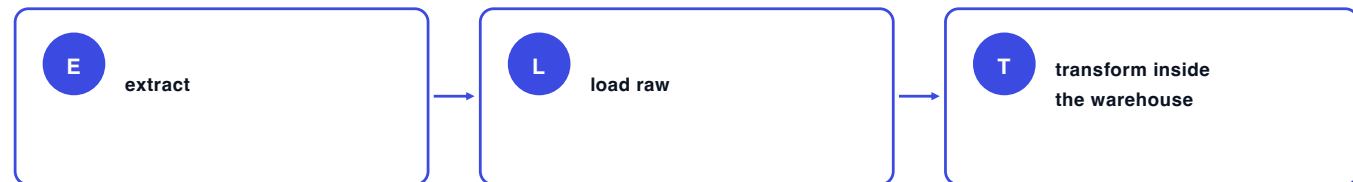
09 **Sort.** Put the rows in an order that makes sense to read.

ETL



The warehouse only ever sees tidy data. The old default, from when warehouse compute was expensive and you rationed it.

ELT



Keeps the raw copy, so a bad transformation can be fixed without going back to the source. The current default.

Same three letters. The only difference is where the T happens, and what you still have if it goes wrong.

— Same three letters. The only difference is where the T happens.

cleaner

GOLD

Shaped for a question, aggregated.

This is what the report reads.

You may read this or you
may build this

SILVER

Cleaned, typed, deduplicated, joined.

One row means one thing.

you may read this

BRONZE

Exactly as it arrived. Unchanged.

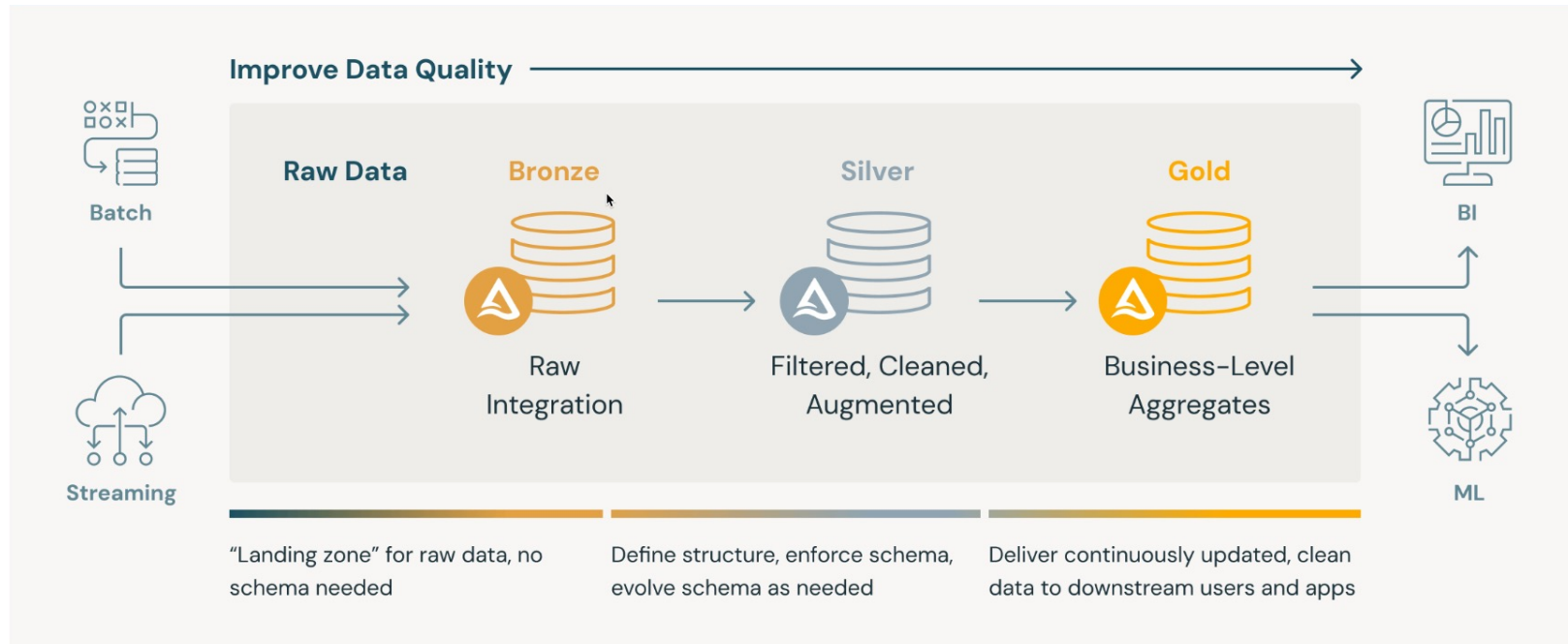
Nobody reports off this, ever.

data engineer

A BI analyst mostly reads silver and builds gold. Bronze belongs to the engineer who gets paged at 3am.

— Bronze as it arrived, silver cleaned, gold shaped for a question.

The term "**medallion architecture**" was coined by **Databricks** to describe a multi-layered design pattern used to logically organize and incrementally improve data quality within a data lakehouse



A BI analyst mostly reads silver and builds gold. Bronze belongs to the engineer who gets paged at 3am.

— Bronze as it arrived, silver cleaned, gold shaped for a question.

Break

 Twenty-five minutes

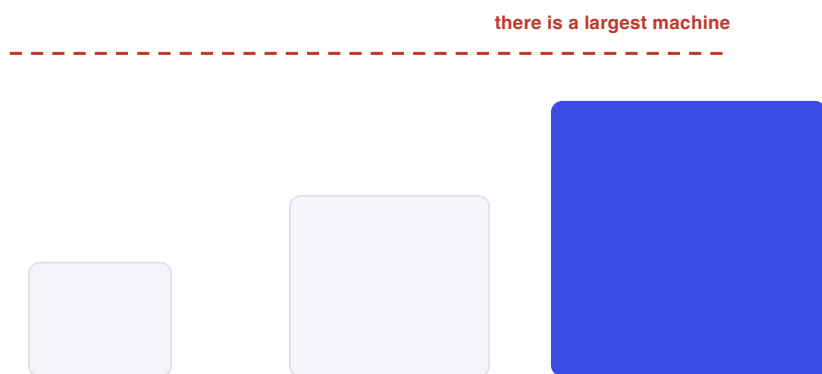
Put questions in the channel and I will pick them up when we come back.

PART 03

Why the cloud warehouse exists

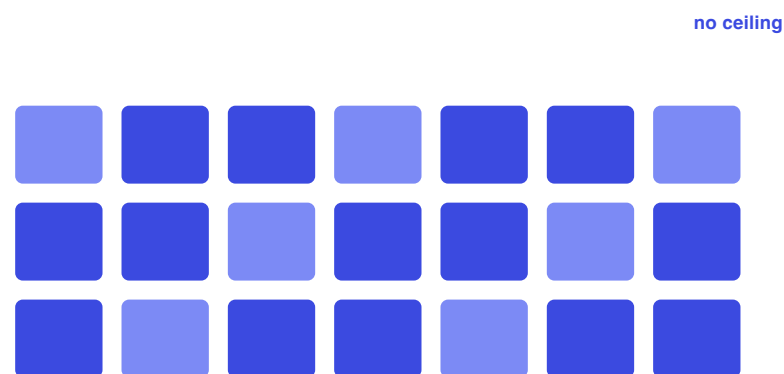
What happened when the data outgrew the machine, and what renting
somebody else's costs you.

Vertical: buy a bigger machine



and the price stops behaving
long before you reach it

Horizontal: use more machines



but the work has to be split, coordinated,
and survive one machine dying mid-job

The cloud warehouse sells you the second one without making you build the hard part.

— Vertical scaling has a ceiling. Horizontal does not, but somebody has to build it.

Why the data went to the cloud

01 You size it on the day,
instead of guessing what
you need in three years
(Elasticity)

02 You stop paying 24/7 for
capacity you only need one
hour a day

03 Somebody else patches it,
backs it up, and replaces
the disk that failed.

04 You start this afternoon,
instead of waiting out a
procurement cycle.

05 It grows with the business,
instead of in five-year
steps **(Scalability)**

06 You plug into the latest
AI/analytics tools the day
they ship, instead of
building and maintaining
them yourself

None of this makes the cloud the right answer everywhere. The next slide is the other side of it.

What you give up

Where your data physically sits

Somebody else's building, often in another country. This is why on-premise is still normal in Jordanian banking, government and healthcare.

A cost you had already paid

On-premise was one large bill, years ago. Cloud is a bill every month, and one careless query can move it.

Working without a connection

No internet, no warehouse. An on-premise server keeps answering on the local network.

Leaving easily

The SQL dialect, the permissions and the pipelines belong to the vendor. Moving off is a project, not a setting.

VENDORS

Who sells a cloud data warehouse

Product	From	What you configure
BigQuery	Google	Nothing. It assigns the machines for you
Snowflake	Snowflake	A warehouse size, which you start and stop
Redshift	Amazon	Historically node types and counts
Synapse and Fabric	Microsoft	Capacity, and where it sits in the tenant

Which one you meet depends on who hires you. The ideas transfer; the buttons do not.

TWO OF THEM

BigQuery and Snowflake

The same idea

Both are managed services you write SQL against.
Both separate storage from compute and bill them
apart. Both scale without you buying hardware.

The buttons differ. What a warehouse is does not.

Different knobs

BigQuery assigns the compute and asks you to
choose nothing. Snowflake gives you a size to pick,
and to start and stop.

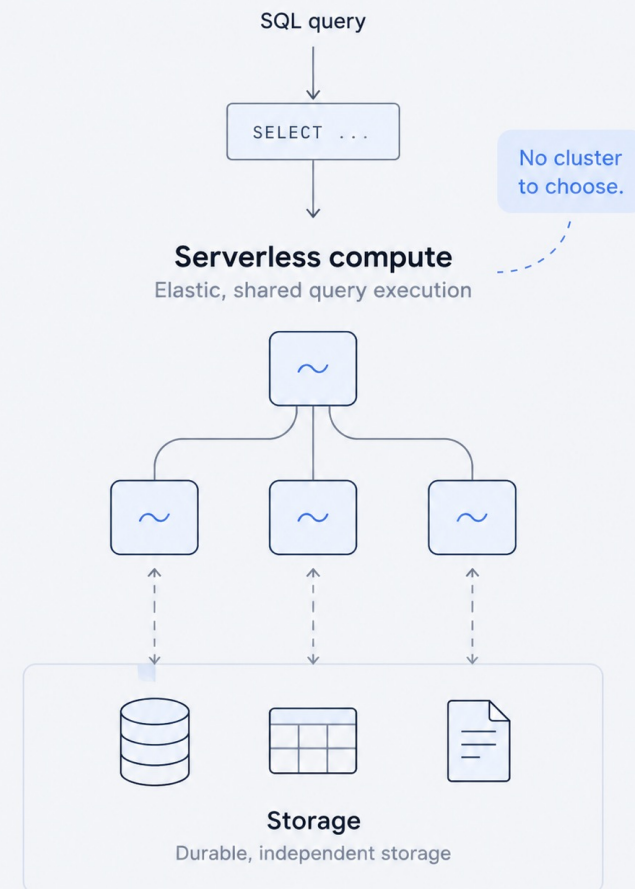
BigQuery runs on Google Cloud. Snowflake runs on
Amazon, Microsoft and Google, which is sometimes
the whole reason it is chosen.

BIGQUERY

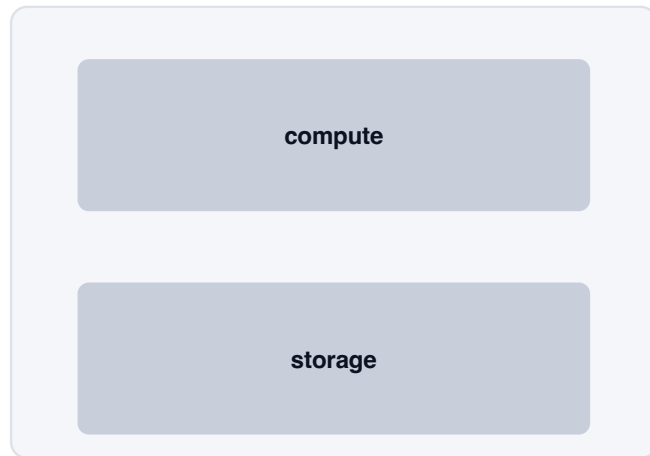
What Google did differently

BigQuery is built on Dremel, an internal Google system published as a paper in 2010. Two of its decisions are the reason your afternoon works: storage and compute were pulled apart, and there is no cluster for you to choose.

Because nothing is reserved for you, nothing has to be paid for while you are not using it. That is what makes a genuinely free sandbox possible, rather than a trial that expires.

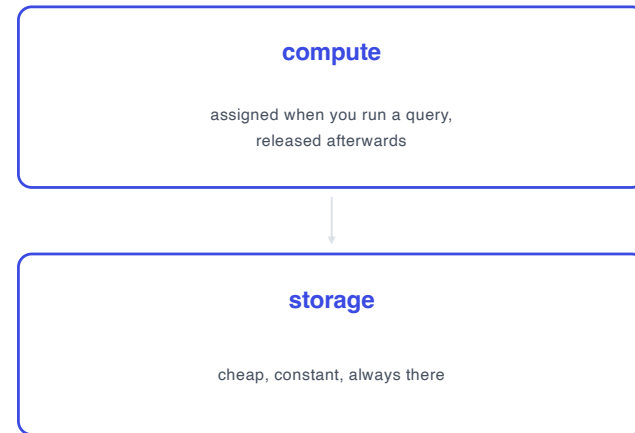


A cluster you size yourself



One box. Paid for whether or not anyone is running a query.

Storage and compute pulled apart



Two bills that move independently. Nothing is reserved for you.

Nothing reserved means nothing to pay for while you are idle. That is what makes a free sandbox possible.

— Storage and compute billed apart is what makes a free sandbox possible.

WHAT YOU SPEND

1 TiB



free every month. Every query spends some of it, and you can see how much before you run it.

BigQuery sandbox documentation, verified 18 September 2026

Before you click anything

01 Region. Where the bytes physically are. It decides speed, it decides which laws apply, and you cannot join across two of them.

02 Permission. Project, then dataset, then table. "Access denied" usually means the level above the one you are staring at.

03 Allowance. 1 TiB of processing a month. 10 GiB of storage for the life of the project, and deleting does not give it back.

Tables expire after 60 days. So export anything you want to keep.

What I am about to do

01 Open the console and pin bigquery-public-data in the Explorer pane.

02 Open the orders table and read the Schema tab.

03 Open Preview and look at real rows, which costs nothing.

04 Open Details and read the row count and the table size.

05 Paste a query in the editor and read the estimate before running it.

Follow along if you can. I will go slowly, and nothing here is hard to catch up on.

Google Cloud Gemini API

← Add data Data sources

Filter By

Data Source Type

- Databases (19)
- Marketing (38)
- Business Applications (30)
- Storage/Data Lakes (6)
- Data Warehouses (3)
- Data Streaming (4)

Most popular data sources

- Google Cloud Storage: Data lake storage and analysis.
- Local File: Data ingestion from local files.

Featured data sources

- Agent Platform: Unified ML platform for building and deploying models.
- Amazon Redshift: Data warehousing and business intelligence.
- Campaign Manager 360: Enterprise ad campaign management and analysis.
- Facebook/Meta Ads: Social media advertising performance and targeting.
- Google Bigtable: NoSQL data analysis for high-throughput applications.
- Google Cloud AlloyDB: Transactional data analysis and reporting.

Public Datasets

- Sharing (Analytics Hub)
- Star a project by name

Marketplace

Find products or ask questions AI Mode

Marketplace > Data

Filter Type to filter 362 results

Category

- AI Agent (1)
- Advertising (26)
- Analytics (58)
- Blockchain (18)
- CRM (4)

Type

- Data

Price

- Free (244)
- Paid (70)

Industry

- Education (10)
- Financial Services and Insurance (1)
- Healthcare and Life sciences (124)

google-books-ngrams-2020
BigQuery Public Data

Google Books word counts found in books from 1500-2012

Data

Solid Date

solid_date is a comprehensive, materialized calendar and date dimension table, enumerating every discrete date value over 500 years from...

Data

Cloud-to-Ground Lightning Strikes
NOAA

Aggregated lightning strike data from 1987 to 2018

Data

Real-time & Historical Data - Global Aircraft Auctions

Data

Public Datasets in the Marketplace

Google Cloud Gemini API

Sandbox Set up billing to upgrade to the full BigQuery experience. [Learn more](#)

Dismiss Upgrade

Search for resources
bigquery-public-data

Home
Starred
Shared with me
Job history

gen-lang-client-0502302409

Datasets
Connections
Conversations
Agents
Queries
(Classic) Queries
Notebooks
Data canvases
Data preparations
Pipelines

Search results for "bigquery-public-data"

Clicking a search result will replace the current tab. To keep these search results open, double-click the *tab name*, or use Cmd+click to open the result in a new tab.

Dismiss Refresh

Project Region Type

Display name	Description	Type	Project	Region	Last modified	Creator	
sample_files Dataset: github_repos	Sampled file metadata for files at HEAD, from the top 400k repositories listed	Table	bigquery-...	us	Jun 28, 201...		⋮ ☆
Variable Dataset: us_census_bureau_acs_5year_via_google_data_commons	Variable is the variable dictionary. Each row defines one statistical variable.	Table	bigquery-...	us	Apr 23, 2026...		⋮ ☆
sample_commits Dataset: github_repos	Sample of [bigquery-public-data:github_repos.commits] table. SELECT *	Table	bigquery-...	us	Jun 24, 201...		⋮ ☆
files Dataset: github_repos	File metadata for all files at HEAD. Join with [bigquery-public-	Table	bigquery-...	us	Nov 27, 202...		⋮ ☆
us_cities_area Dataset: utility_us	This dataset has moved! You can find all of the datasets that define the	Table	bigquery-...	us	Mar 19, 201...		⋮ ☆
Transactions Dataset: crypto_solana_mainnet_us	-	Table	bigquery-...	us-centra...	Sep 3, 2026...		⋮ ☆
order_items Dataset: thelook_ecommerce	Programmatically generated order items for The Look fictitious e-commerce store	Table	bigquery-...	us	Sep 21, 202...		⋮ ☆
posts_questions Dataset: stackoverflow	-	Table	bigquery-...	us	Nov 25, 202...		⋮ ☆
states Dataset: geo_us_boundaries	-	Table	bigquery-...	us	Nov 20, 201...		⋮ ☆
wikipedia Dataset: samples	-	Table	bigquery-...	us	Aug 28, 202...		⋮ ☆
logs Dataset: crypto_ethereum	Similar to the token_transfers table, the logs table contains data for smart	Table	bigquery-...	us	Nov 23, 202...		⋮ ☆
contracts Dataset: crypto_ethereum	Some transactions create smart contracts from their input bytes, and this	Table	bigquery-...	us	Nov 23, 202...		⋮ ☆
ghcnd_2016 Dataset: ...	-	Table	bigquery-...	us	Dec 26, 202...		⋮ ☆

The Explorer pane. bigquery-public-data is somebody else's project, and you are allowed to read it.

Google Cloud Gemini API

Sandbox Set up billing to upgrade to the full BigQuery experience. [Learn more](#) Dismiss Upgrade

Untitled query Search

Explorer + Add data

Search for resources y-public-data.thelook_ecommerce X

Home
Starred
Shared with me
Job history

gen-lang-client-0502302409

Datasets
Connections
Conversations
Agents
Queries
(Classic) Queries
Notebooks
Data canvases
Data preparations
Pipelines

Search results for "bigquery-public-data.thelook_ecommerce"

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Project Region Type

Display name	Description	Type	Project	Region	Last modified	Creator	
order_items Dataset: thelook_ecommerce	Programmatically generated order items for The Look fictitious e-commerce store	Table	bigquery-public-data	us	Sep 21, 2026, 6:02:46 AM UTC+3		: ☆
products Dataset: thelook_ecommerce	The Look fictitious e-commerce dataset - products table	Table	bigquery-public-data	us	Sep 21, 2026, 6:02:46 AM UTC+3		: ☆
users Dataset: thelook_ecommerce	Programmatically generated users for The Look fictitious e-commerce store	Table	bigquery-public-data	us	Sep 21, 2026, 6:03:07 AM UTC+3		: ☆
inventory_items Dataset: thelook_ecommerce	Programmatically generated inventory for The Look fictitious e-commerce store	Table	bigquery-public-data	us	Sep 21, 2026, 6:02:53 AM UTC+3		: ☆
orders Dataset: thelook_ecommerce	Programmatically generated orders for The Look fictitious e-commerce store	Table	bigquery-public-data	us	Sep 21, 2026, 6:02:47 AM UTC+3		: ☆
events Dataset: thelook_ecommerce	Programmatically generated web events for The Look fictitious e-commerce store	Table	bigquery-public-data	us	Sep 21, 2026, 6:02:54 AM UTC+3		: ☆
distribution_centers Dataset: thelook_ecommerce	The Look fictitious e-commerce dataset: distribution_centers table	Table	bigquery-public-data	us	Sep 21, 2026, 6:03:15 AM UTC+3		: ☆
thelook_ecommerce	Fictitious E-Commerce Dataset	Dataset	bigquery-public-data	us	Mar 9, 2026, 5:08:04 PM UTC+3		: ☆
graph Dataset: thelook_ecommerce	-	Graph	bigquery-public-data	us	Apr 23, 2026, 10:02:00 AM UTC+3		: ☆

The Explorer pane. bigquery-public-data is somebody else's project, and you are allowed to read it.

Schema										
Details										
Preview										
Insights										
Lineage										
Data Profile										
Data Quality										
Filter Enter property name or value ?										
☐	Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags ?	Data Policies
☐	order_id	INTEGER	NULLABLE	-	-	-	-	-	-	-
☐	user_id	INTEGER	NULLABLE	-	-	-	-	-	-	-
☐	status	STRING	NULLABLE	-	-	-	-	-	-	-
☐	gender	STRING	NULLABLE	-	-	-	-	-	-	-
☐	created_at	TIMESTAMP	NULLABLE	-	-	-	-	-	-	-
☐	returned_at	TIMESTAMP	NULLABLE	-	-	-	-	-	-	-
☐	shipped_at	TIMESTAMP	NULLABLE	-	-	-	-	-	-	-
☐	delivered_at	TIMESTAMP	NULLABLE	-	-	-	-	-	-	-
☐	num_of_item	INTEGER	NULLABLE	-	-	-	-	-	-	-

— The Schema tab. Every column, its type, and whether it is allowed to be empty.

Schema

Details

Preview

Insights

Lineage

Data Profile

Data Quality

Row	order_id	user_id	status	gender	created_at	returned_at	shipped_at	delivered_at	num_of_
1	8	9	Cancelled	F	2025-11-13 20:25:11 UTC	null	null	null	
2	44	36	Cancelled	F	2025-09-13 16:34:13 UTC	null	null	null	
3	71	52	Cancelled	F	2019-10-28 01:10:23 UTC	null	null	null	
4	80	60	Cancelled	F	2022-07-25 07:57:50 UTC	null	null	null	
5	101	79	Cancelled	F	2024-01-11 06:10:44 UTC	null	null	null	
6	102	82	Cancelled	F	2021-12-02 11:32:36 UTC	null	null	null	
7	103	83	Cancelled	F	2026-05-22 21:46:17 UTC	null	null	null	
8	109	90	Cancelled	F	2026-08-11 00:35:02 UTC	null	null	null	
9	110	90	Cancelled	F	2026-08-28 11:46:27 UTC	null	null	null	
10	113	91	Cancelled	F	2023-08-24 23:35:21 UTC	null	null	null	
11	121	97	Cancelled	F	2021-09-03 23:01:49 UTC	null	null	null	
12	152	125	Cancelled	F	2025-09-10 06:46:07 UTC	null	null	null	
13	162	132	Cancelled	F	2025-04-05 03:33:06 UTC	null	null	null	
14	177	142	Cancelled	F	2022-05-14 11:52:50 UTC	null	null	null	
15	180	144	Cancelled	F	2022-07-25 02:08:02 UTC	null	null	null	
16	182	145	Cancelled	F	2024-04-29 12:10:59 UTC	null	null	null	
17	185	148	Cancelled	F	2024-10-18 12:23:11 UTC	null	null	null	
18	187	154	Cancelled	F	2026-03-29 09:10:35 UTC	null	null	null	
19	194	158	Cancelled	F	2025-03-29 13:32:24 UTC	null	null	null	
20	196	158	Cancelled	F	2024-03-22 01:16:05 UTC	null	null	null	
21	204	164	Cancelled	F	2023-09-01 14:15:37 UTC	null	null	null	
22	220	178	Cancelled	F	2025-04-28 03:29:19 UTC	null	null	null	
23	264	216	Cancelled	F	2025-03-02 02:42:26 UTC	null	null	null	

Results per page:

50

1 – 50 of 124942

Results per page: 50 1 - 50 of 124942

Preview. Real rows, no query, no cost. This is the first thing to open, always.

Table info

[Edit Details](#)

Table ID	bigquery-public-data.thelook_ecommerce.orders 
Created	Feb 26, 2022, 2:52:03 AM UTC+3
Last modified	Sep 21, 2026, 6:02:47 AM UTC+3
Table expiration	NEVER
Data location	US
Default collation	
Default rounding mode	ROUNDING_MODE_UNSPECIFIED
Case insensitive	false
Description	Programatically generated orders for The Look fictitious e-commerce store
Labels	
Primary key(s)	
Tags	

Storage info

Number of rows	124,942
Total logical bytes	6.44 MB
Active logical bytes	6.44 MB
Long term logical bytes	0 B
Current physical bytes	2.11 MB
Total physical bytes	16.88 MB
Active physical bytes	16.88 MB
Long term physical	0 B

Details. The row count and the size of the table, which the next slide compares against.

Google Cloud Gemini API Search (/) for resources, docs, products, and more

Sandbox Set up billing to upgrade to the full BigQuery experience. [Learn more](#) Dismiss Upgrade

Explorer + Add data

Search for resources

- Home
- Starred
- Shared with me
- Job history
- gen-lang-client-0502302409
 - Datasets
 - Connections
 - Conversations
 - Agents
 - Queries
 - (Classic) Queries
 - Notebooks
 - Data canvases
 - Data preparations
 - Pipelines

Untitled query Run Save Schedule Share

File Edit Run Tools

```
1 select * from `bigquery-public-data.thelook_ecommerce.orders` limit 10;
```

Query completed
Using on-demand processing quota

Recent 11:41 AM

Query results Chat View all jobs Save results Open in

Job information Results Visualization JSON Execution details Execution graph

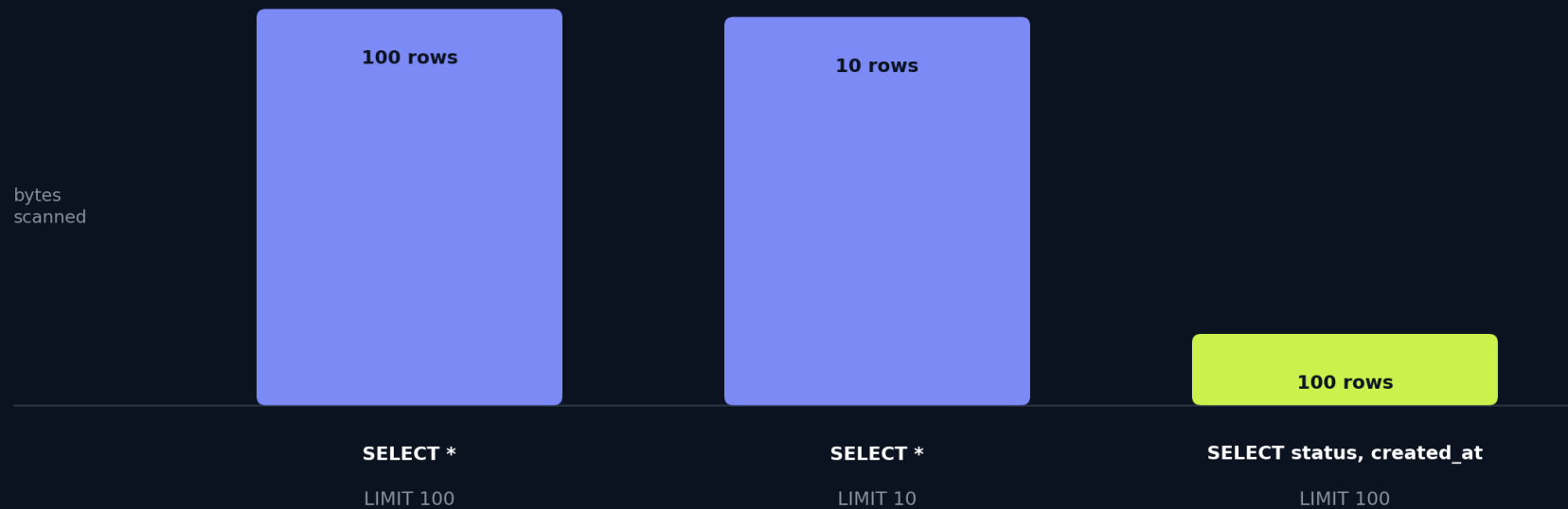
Row	order_id	user_id	status	gender	created_at	returned_at	shipped_at
1	8	9	Cancelled	F	2025-11-13 20:25:11 UTC	null	null
2	44	36	Cancelled	F	2025-09-13 16:34:13 UTC	null	null
3	71	52	Cancelled	F	2019-10-28 01:10:23 UTC	null	null
4	80	60	Cancelled	F	2022-07-25 07:57:50 UTC	null	null
5	101	79	Cancelled	F	2024-01-11 06:10:44 UTC	null	null
6	102	82	Cancelled	F	2021-12-02 11:32:36 UTC	null	null
7	103	83	Cancelled	F	2026-05-22 21:46:17 UTC	null	null
8	109	90	Cancelled	F	2026-08-11 00:35:02 UTC	null	null
9	110	90	Cancelled	F	2026-08-28 11:46:27 UTC	null	null
10	113	91	Cancelled	F	2023-08-24 23:35:21 UTC	null	null

The editor, before you press run. The validator tells you what the query will process.

Same table. Three queries.

PLACEHOLDER HEIGHTS. Measure in rehearsal, edit VALUES in make_d02_figures.py, re-run.

LIMIT changed the rows returned. It barely changed the bytes read.



Ask for fewer columns and you read less. Ask for fewer rows and, on this table, you do not.

— LIMIT changed the rows returned. It barely changed the bytes read.

THE QUERY

The query, in plain English

read it, do not write it

```
SELECT  status, COUNT(*) AS orders
        -- what I want back: each status, and how many orders have it
FROM    `bigquery-public-data.thelook_ecommerce.orders`
        -- which table it comes from
WHERE   created_at >= '2024-01-01'
        -- which rows to keep
GROUP BY status
        -- fold the rows together so there is one line per status
```

Read it, run it, and do not change it. Writing queries starts on Sunday, from nothing.

YOUR
TURN

13 minutes

First look at a real table

TASK

Open the orders table. Read the Schema and the Preview. Then paste the query, read the estimate, and run it.

NO ACCOUNT YET

Use the supplied results file and the screenshots. You reach the same two checkpoints and you are not behind.

DONE WHEN

You can say how many columns the table has, what one row of your result means, and what the query processed.

Access denied means check you picked a project. No rows back means check the date line is exactly as printed.

TOGETHER

30 minutes

The whole pipeline, on one page

TASK

Mark up your sheet as I build the diagram. Write the name of each box and who owns it.

THEN

We go back across it a second time and mark every place SQL shows up. There are four.

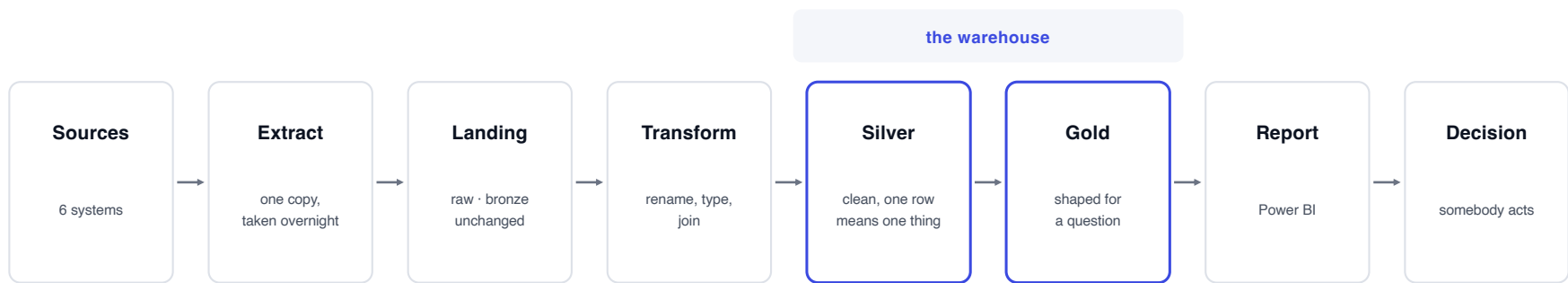
KEEP IT

This sheet is the map for the next six weeks. Everything after today is one box on it.

Eight boxes. We fill them in together.



— Eight boxes. We fill them in together, left to right.



Everything you learn after today is one box on this line.

— The finished pipeline. Everything after today is one box on this line.



One language, four places on the same line. That is why SQL comes next.

— The same line again, with every place SQL appears marked on it.

Four places SQL shows up

At the source

A SELECT is how the copy gets pulled out of the operational database in the first place.

Over the landing zone

You can query files where they sit, without loading them first, using external tables.

Inside the warehouse

In an ELT pipeline, SQL is what does the transformation. Silver and gold are built with it.

At the reporting layer

SQL sits behind Power BI Get Data and DirectQuery, and custom SQL is an option on most connections.

RECAP

What you can do now

- Name where data lives, and what shape it is in
- Explain why we copy data out of the source
- Tell an operational system from an analytical one
- Read what a query will cost before you run it

Questions?

 Next session: Sun 27 Sep

D03 is SQL, from the beginning, with no prior knowledge assumed. Your three tools must work by then.